Mid-Semester Exam - Deferred

EGH445 Modern Control - 2026

This exam has a total of 3 problems.
The total number of possible marks for this exam is 100.

It is recommended that you attempt to answer all questions, with each response carefully justified. Explanations of the steps used are essential, and will be accounted for in the assessment.

- **Full marks** are awarded to **fully justified and numerically correct** solutions.
- **Partial marks** are awarded to **completed but numerically incorrect** solutions, **partially complete** solutions, or solutions that have used **clearly stated and justified assumptions** consistent with the question.

Problem 1

Provide a **short response** (2-3 sentences) to the following questions (a) - (h). You may include supporting diagrams, mathematics and examples (real or theoretical) in your responses.

(a) Describe what a system *state* and a *redundant state* represent in terms of real dynamic systems.

(b) Describe what equilibrium points (E.P.) are for a dynamic system and explain, with an example, how multiple equilibrium points can arise in a nonlinear system.

(c) Consider the system response from a linearised discrete time dynamic system (i) and two different pole (x) zero (o) plots (ii) and (iii) in Figure 1. Which, if any, of the pole zero plots would you expect to correspond to the system response? Justify your response.

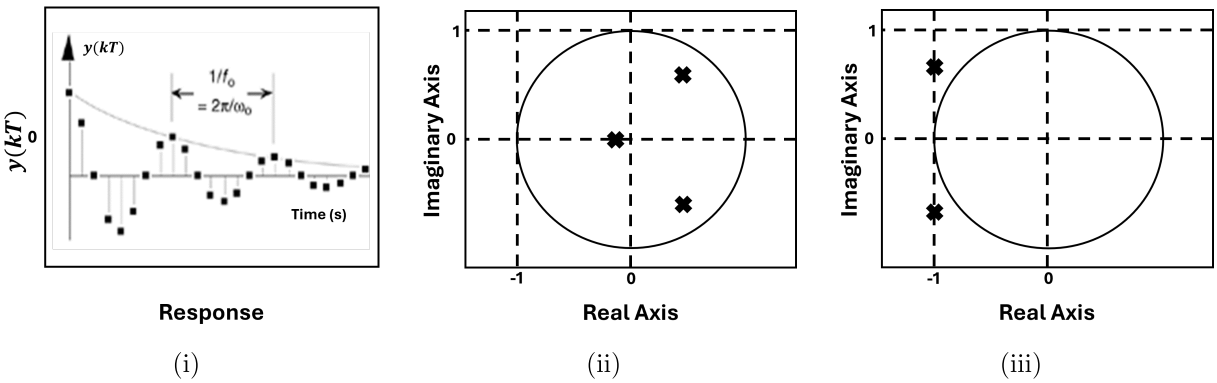

Figure 1: System Response and Pole-Zero Plots

(d) If a linear time-invariant (LTI) system is asymptotically stable, is it internally stable, bounded-input bounded-output stable, or both? Explain your reasoning.

(e) What is the purpose of a linear transformation matrix P in state feedback control system design?

(f) What is a Lyapunov function used for in control system analysis and design and if such function is decreasing in value, what does this mean? Explain your reasoning.

(30 marks)

Problem 2

Consider an insulin-glucose model representing part of the endocrine system where glucose stimulates insulin secretion, insulin lowers blood glucose, and external insulin infusion helps regulate the system. The system is complex, but can be modelled using simplified dynamics given by (1) - (2) such that

$$\dot{x}_1(t) = -a\,x_1(t)x_2(t) + \frac{b}{x_1(t)} \tag{1}$$

$$\dot{x}_2(t) = -c\,x_2(t) + k\sin(x_2(t)) + u(t) \tag{2}$$

where a, b, c and k and constants, $x_1(t)$ is the blood glucose concentration (mmol/L), $x_2(t)$ is the insulin concentration (μU/L) and $u(t)$ is the external insulin injection rate (U/min) via an artificial pancreas controller. The unit U is a clinical measurement of insulin units, mmol is milli moles (concentration) and L is litres.

(a) Linearise the nonlinear system (1) - (2) for a general (non-specific) equilibrium point and express the result in the standard linear state-space form: $\delta\dot{x} = A\delta x + B\delta u$, clearly stating A and B.

(b) Analyse the stability of the linear and non-linear systems, stating conditions for stability considering physically meaningful state and parameter values (i.e. values that are physically realisable or practical).

(40 marks)

Problem 3

Consider a simple financial system representing the current real estate market in Australia, where investor speculation, mortgage stress and affordability are balanced by policy and rules. The system can be modelled using simplified dynamics given by (3) - (5) such that

$$\dot{x}_1(t) = \alpha\,x_1(t) + b_1\,u(t) \tag{3}$$

$$\dot{x}_2(t) = \beta\,x_2(t) + b_2\,u(t) \tag{4}$$

$$y(t) = x_1(t) \tag{5}$$

where α, β, b_1 and b_2 and constants, $x_1(t)$ is investor speculation (e.g. FOMO), $x_2(t)$ is the mortgage stress (affordability) and $u(t)$ is the reserve bank interest rate proxy implemented via policy and rules.

(a) Derive a minimum order state-space model for the system given in (3) - (5) and simplify assuming that $\alpha = 2$, $\beta = -1$, $b_1 = 1$ and $b_2 = 3$.

(b) A discrete time state space model for the continuous system in (a) with sampling time $T = 1$ year is given in (6) - (7). Find the missing values a, b and c for the discrete time model, showing your working.

$$x(kT+T) = \begin{bmatrix} a & b \\ b & c \end{bmatrix} x(kT) + \begin{bmatrix} 3.195 \\ 1.896 \end{bmatrix} u(kT) \tag{6}$$

$$y(kT) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(kT) \tag{7}$$

(30 marks)

(c) A state feedback controller $u(kT) = -Kx(kT)$ is designed for the discrete time system (6) - (7) such that $K = [2.466\ \text{-}0.015]$. Analyse the expected closed loop system performance and if possible provide a practical interpretation.

[***Note:*** *If you are unable to solve (b), assume* $a = 7.5$, $b = 0$ *and* $c = 0.5$]

END OF EXAM

RESOURCE SHEET

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$
1.	–	–	Kronecker delta $\delta_0(k)$ 1 $\quad k = 0$ 0 $\quad k \neq 0$	1
2.	–	–	$\delta_0(n-k)$ 1 $\quad n = k$ 0 $\quad n \neq k$	z^{-k}
3.	$\frac{1}{s}$	$1(t)$	$1(k)$	$\frac{1}{1-z^{-1}}$
4.	$\frac{1}{s+a}$	e^{-at}	e^{-akT}	$\frac{1}{1-e^{-aT}z^{-1}}$
5.	$\frac{1}{s^2}$	t	kT	$\frac{Tz^{-1}}{(1-z^{-1})^2}$
6.	$\frac{2}{s^3}$	t^2	$(kT)^2$	$\frac{T^2 z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$
7.	$\frac{6}{s^4}$	t^3	$(kT)^3$	$\frac{T^3 z^{-1}(1+4z^{-1}+z^{-2})}{(1-z^{-1})^4}$
8.	$\frac{a}{s(s+a)}$	$1 - e^{-at}$	$1 - e^{-akT}$	$\frac{(1-e^{-aT})z^{-1}}{(1-z^{-1})(1-e^{-aT}z^{-1})}$
9.	$\frac{b-a}{(s+a)(s+b)}$	$e^{-at} - e^{-bt}$	$e^{-akT} - e^{-bkT}$	$\frac{(e^{-aT}-e^{-bT})z^{-1}}{(1-e^{-aT}z^{-1})(1-e^{-bT}z^{-1})}$
10.	$\frac{1}{(s+a)^2}$	te^{-at}	kTe^{-akT}	$\frac{Te^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
11.	$\frac{s}{(s+a)^2}$	$(1-at)e^{-at}$	$(1-akT)e^{-akT}$	$\frac{1-(1+aT)e^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
12.	$\frac{2}{(s+a)^3}$	t^2e^{-at}	$(kT)^2e^{-akT}$	$\frac{T^2e^{-aT}(1+e^{-aT}z^{-1})z^{-1}}{(1-e^{-aT}z^{-1})^3}$
13.	$\frac{a^2}{s^2(s+a)}$	$at - 1 + e^{-at}$	$akT - 1 + e^{-akT}$	$\frac{[(aT-1+e^{-aT})+(1-e^{-aT}-aTe^{-aT})z^{-1}]z^{-1}}{(1-z^{-1})^2(1-e^{-aT}z^{-1})}$
14.	$\frac{\omega}{s^2+\omega^2}$	$\sin \omega t$	$\sin \omega kT$	$\frac{z^{-1}\sin \omega T}{1-2z^{-1}\cos \omega T + z^{-2}}$
15.	$\frac{s}{s^2+\omega^2}$	$\cos \omega t$	$\cos \omega kT$	$\frac{1-z^{-1}\cos \omega T}{1-2z^{-1}\cos \omega T + z^{-2}}$
16.	$\frac{\omega}{(s+a)^2+\omega^2}$	$e^{-at}\sin \omega t$	$e^{-akT}\sin \omega kT$	$\frac{e^{-aT}z^{-1}\sin \omega T}{1-2e^{-aT}z^{-1}\cos \omega T + e^{-2aT}z^{-2}}$
17.	$\frac{s+a}{(s+a)^2+\omega^2}$	$e^{-at}\cos \omega t$	$e^{-akT}\cos \omega kT$	$\frac{1-e^{-aT}z^{-1}\cos \omega T}{1-2e^{-aT}z^{-1}\cos \omega T + e^{-2aT}z^{-2}}$
18.	–	–	a^k	$\frac{1}{1-az^{-1}}$
19.	–	–	a^{k-1} $k = 1, 2, 3, \ldots$	$\frac{z^{-1}}{1-az^{-1}}$
20.	–	–	ka^{k-1}	$\frac{z^{-1}}{(1-az^{-1})^2}$
21.	–	–	k^2a^{k-1}	$\frac{z^{-1}(1+az^{-1})}{(1-az^{-1})^3}$
22.	–	–	k^3a^{k-1}	$\frac{z^{-1}(1+4az^{-1}+a^2z^{-2})}{(1-az^{-1})^4}$
23.	–	–	k^4a^{k-1}	$\frac{z^{-1}(1+11az^{-1}+11a^2z^{-2}+a^3z^{-3})}{(1-az^{-1})^5}$
24.	–	–	$a^k \cos k\pi$	$\frac{1}{1+az^{-1}}$

$x(t) = 0$ for $t < 0$
$x(kT) = x(k) = 0$ for $k < 0$
Unless otherwise noted, $k = 0, 1, 2, 3, \ldots$